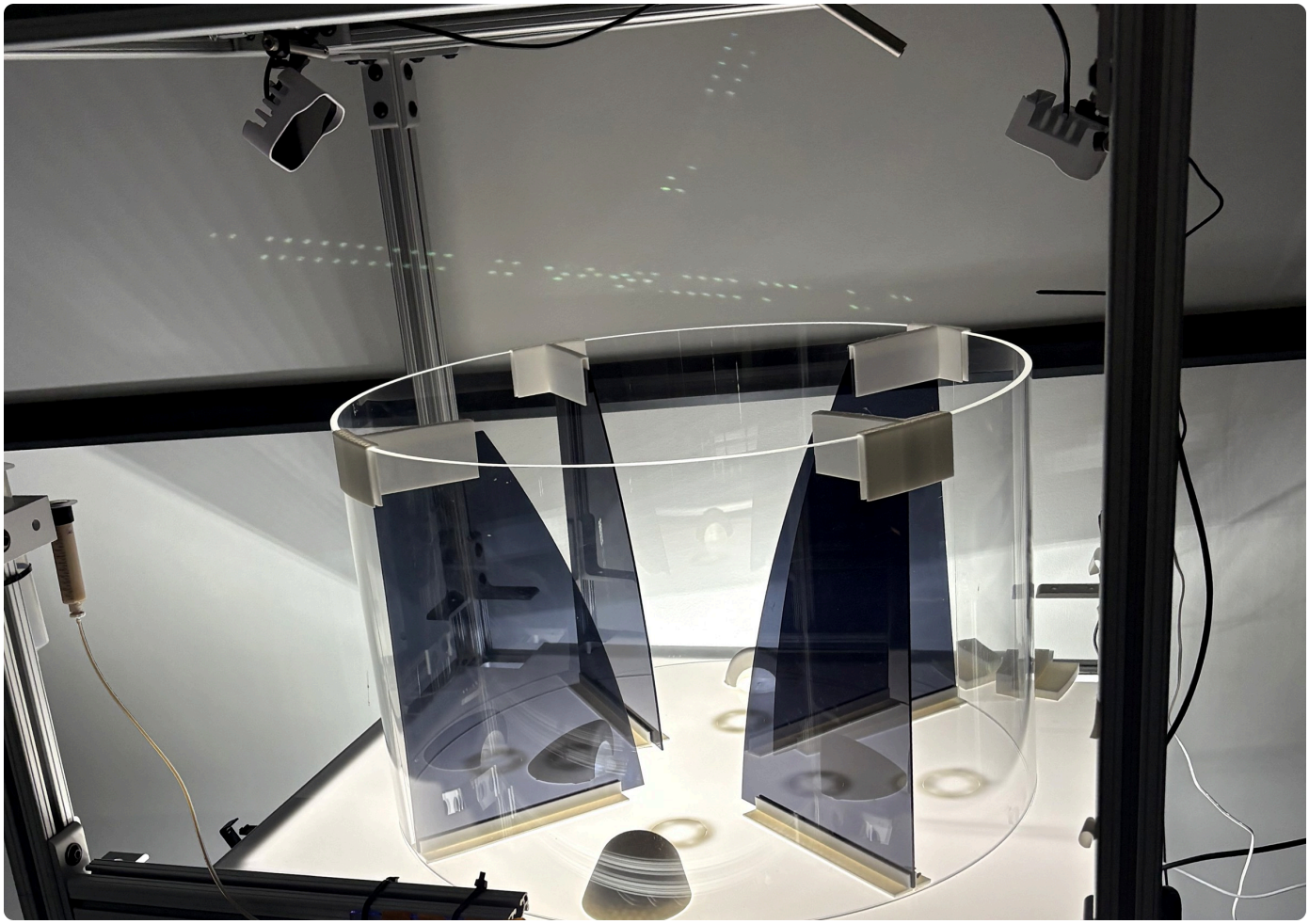
Bora Mandic

Mechanical Design Portfolio · Stanford University · boram@stanford.edu



Behavioral rig I designed and built: aluminum-extrusion frame, floor ports, and arena wall holders.

ABOUT

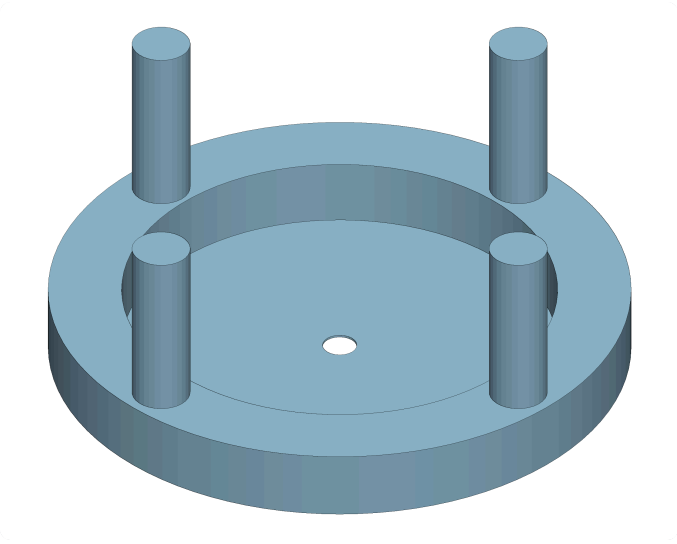
I'm a Stanford mechanical engineering student interested in biotech and neurotech, designing parts for neuroscience and electrical engineering research: a full behavioral rig, a leaf sensor mount, a head mount, and lab fixtures. Every design here is parametric in Fusion, so parts can be quickly adjusted per experiment or per mouse, and each was printed, tested, and refined in the lab.

SELECTED WORK

- Behavioral rig: frame, arena port system (3 design generations, 24 versions), wall holders
- Leaf sensor mount (S4 Lab): 6 versions, ongoing
- Circular head mount: 9 versions
- Lab fixtures: surgery mount holder, plating aid

TOOLS

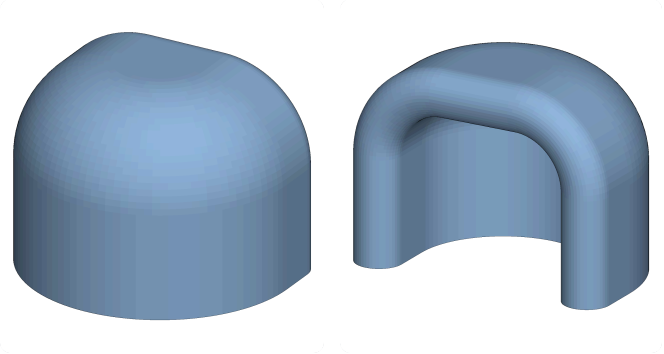
Autodesk Fusion (parametric CAD) · FDM printing (Bambu, Prusa) · laser cutting · Arduino · MATLAB





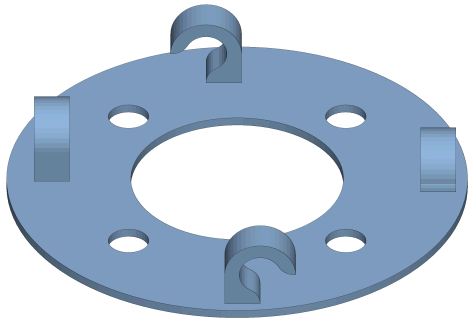
Clear Interlocking Port

Reward port set into the arena floor. Printed clear so the backlit floor shines through and the camera below can capture licks. Locks into the interlocking base.



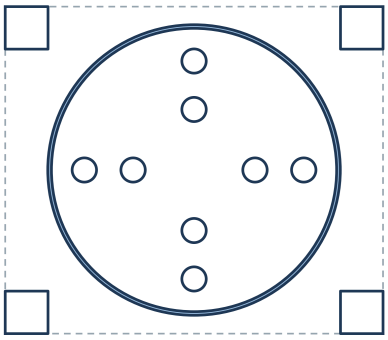
Domed Port

Dome over the port keeps mice from walking across the ports, and shapes the light for the base camera so licks are captured more reliably.



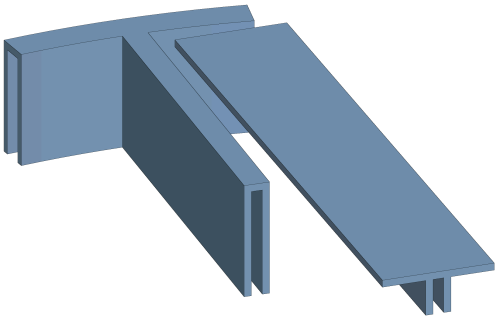
Interlocking Ports Base

Glued to the arena floor. Holes align each port with the floor openings, and built-in hooks route and hold the tubing.



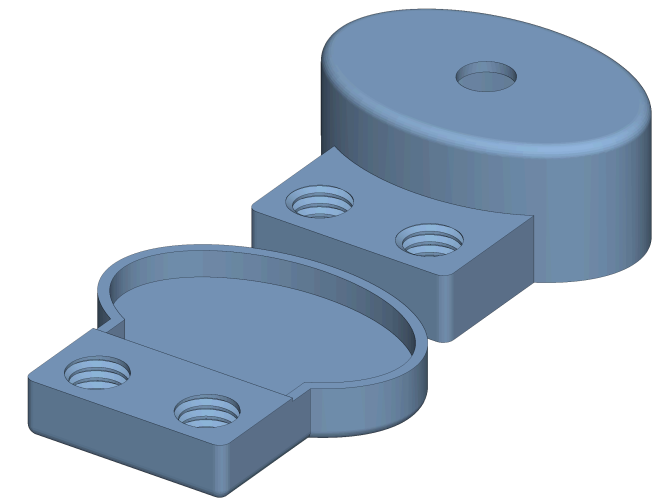
Laser-Cut Arena Floor

Six revisions. An engraved 24-inch ring (shaded) seats the acrylic arena wall, and eight port openings in a cross pattern line up with the interlocking bases.

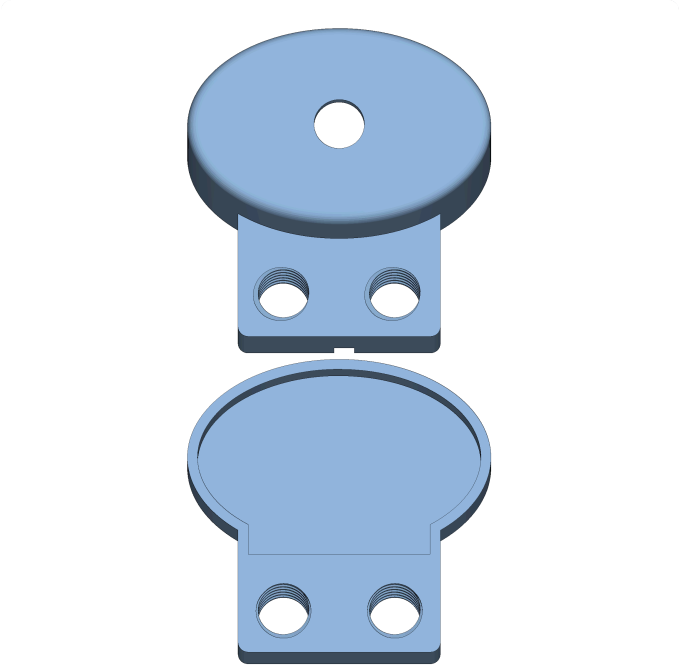


Arena Wall Holders

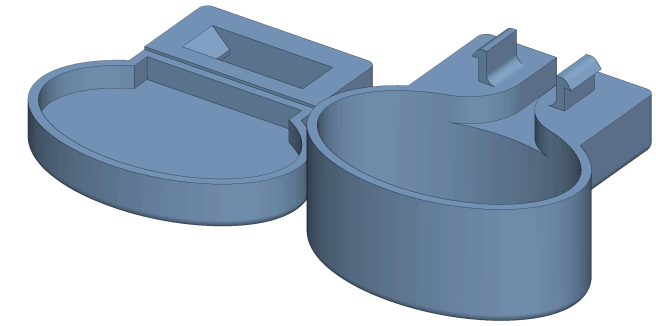
Hold the acrylic arena walls rigid so they don't bend or give way, even with mice climbing and pushing on them.



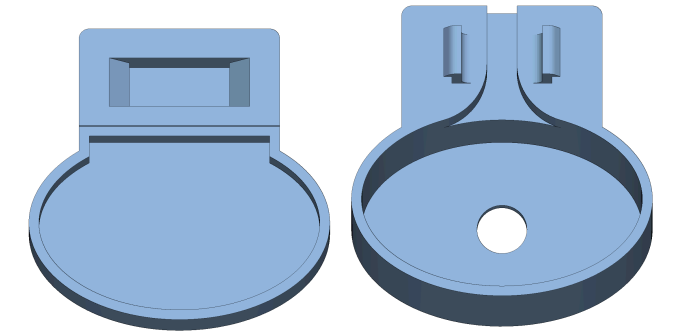
v3: screwed on



v3: top cap outer face (upper), other half's interior (lower)



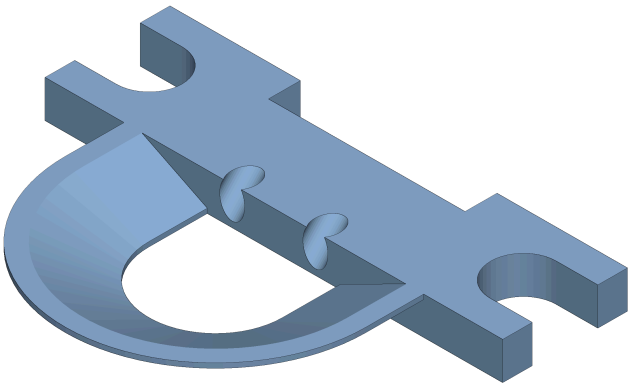
v6: built-in clip

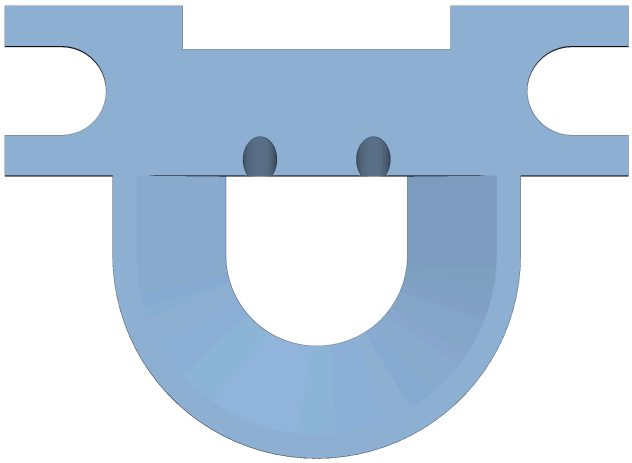


v6: interiors of both halves

Leaf Sensor Mount

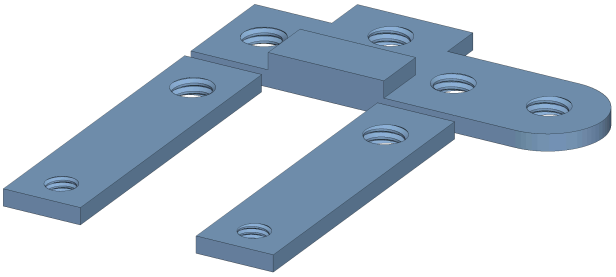
For the S4 Lab: a mount that clips onto the side of a leaf, dimensioned around our specific sensor. The inside is filled with foam, which pushes the sensor against the leaf. Earlier versions screwed on; I built in a clip instead, so no additional hardware or weight is needed. Six versions so far, and the work is ongoing.





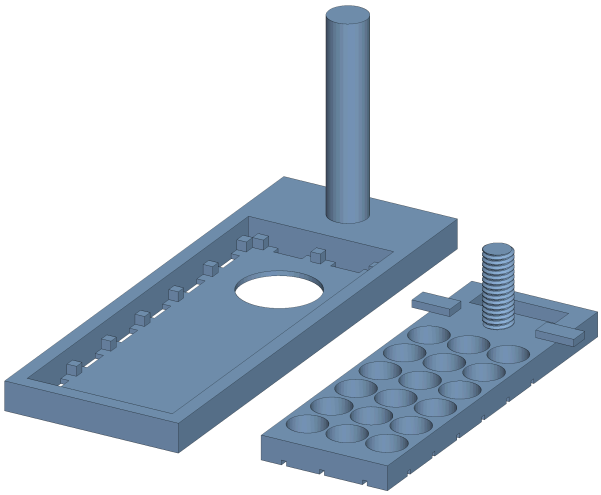
Circular Head Mount

Holds the mouse's head steady on the rig. The ring surrounds the surgical site, which makes surgery easier, and forms a well that keeps saline in place between behavior sessions, improving animal welfare. Includes two ports, one for a fiber optic and one for a grounding pin.



Surgery Mount Holder

Custom-fit to our surgery setup. It holds the head mount steady while it's attached, and several attachment options keep the work area in view during placement.



Parametric Plating Aid

Adjustable wells for plating different brain slices onto microscope slides. It made my plating faster and cleaner; I now out-plate a postdoc with over a decade of experience.